Sales and Profitability Analysis

Sales, profit and discounts in a Superstore dataset, 2011 to 2014

Tools	Microsoft Excel 365, PostgreSQL / pgAdmin, Tableau Public
Dataset	Superstore order-level dataset, 51,290 records, 2011 to 2014
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1. Summary

For this project I looked at sales and profit in a Superstore dataset of 51,290 orders from 2011 to 2014. I checked the data in Excel first, used SQL in PostgreSQL to work out the numbers, and built a dashboard in Tableau Public to show the results by product category.

Total sales came to about 12.64 million and total profit to about 1.47 million, which is an overall profit margin of 11.62%. Technology had the highest sales and profit. Furniture had the lowest margin of the three categories, at 6.98%. Among the subcategories, Tables was the only one that lost money overall, about 64,083.39.

I also looked at Furniture by discount level. Margins got worse as discounts got bigger, and the biggest discounts were losing money. That doesn't prove the discounts caused the losses, because product costs, product mix and order size could also play a part.

2. The question

The question I wanted to answer was:

> *How does sales performance compare with profitability across product categories, and where could the business look to improve profit?*

To get there, I wanted to:

- summarise total sales, profit, quantity and profit margin
- compare sales, profit and margin across the three product categories
- find subcategories that did especially well or badly
- see how Furniture margins change at different discount levels
- write up recommendations and be clear about what the data can and can't show

3. Data and tools

The data is the Superstore order-level dataset, which I worked with as a CSV file. It has order and shipping dates, customer and market details, product category and subcategory, sales, quantity, discount, profit and shipping cost, covering 2011 to 2014.

Tool	What I used it for
Excel 365	Checking the data for blanks and duplicates, converting dates in a separate column, and some quick exploratory checks.
PostgreSQL / pgAdmin	The calculations: KPIs, category and subcategory comparisons, and the Furniture discount analysis.
Tableau Public	A published dashboard comparing sales, profit and profit margin by category, with a written key finding.

4. Method

1. In Excel, I checked the key fields for blanks and looked for exact duplicate rows. There were none of either. I also added a converted order date column and checked that it read as a date.
2. In PostgreSQL, the sales, profit and quantity fields had been imported as text because some values contain commas. I removed the commas and cast the values to numbers before adding anything up.
3. I calculated the record count, total sales, total profit, total quantity, overall profit margin and the number of zero-profit records.
4. I added up sales and profit by category and by subcategory, then worked out margin as profit divided by sales.
5. I compared profit and margin across the discount levels on Furniture orders.
6. I built the Tableau Public dashboard with three charts (sales, profit and profit margin by category) and a short key finding.

5. Overall numbers

Metric	Result
Total records	51,290
Total sales	12,642,905.00
Total profit	1,469,034.82
Total quantity	178,312
Overall profit margin	11.62%
Zero-profit records	668

Profit margin is total profit divided by total sales, multiplied by 100. Figures are rounded.

6. Results

6.1 Categories

Category	Sales	Profit	Profit margin
Furniture	4,110,884	286,782.25	6.98%
Office Supplies	3,787,330	518,473.83	13.69%
Technology	4,744,691	663,778.73	13.99%

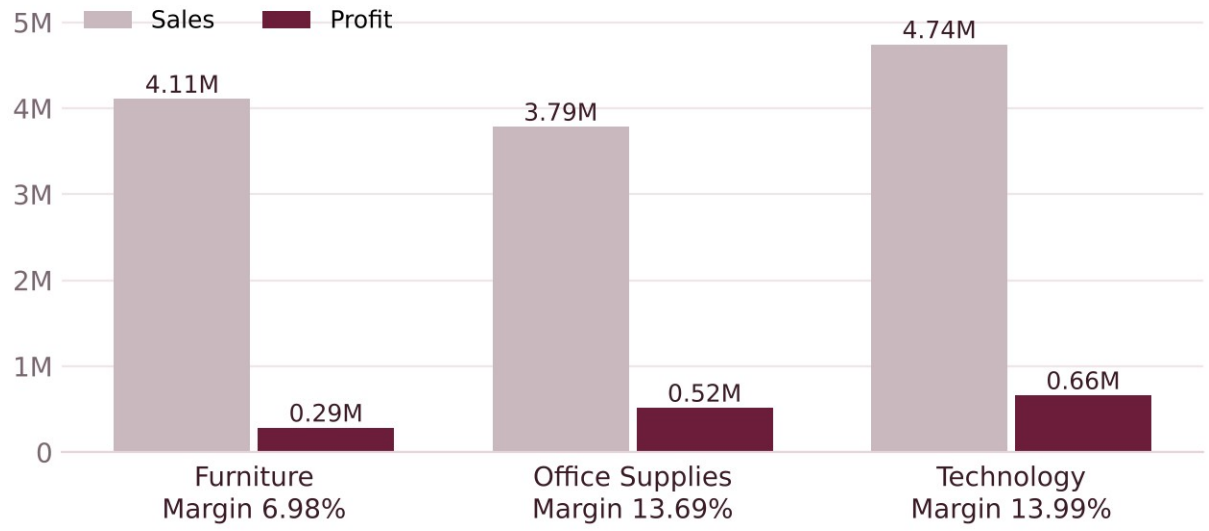


Figure 1. Sales and profit by product category

Technology had the highest sales, profit and margin (13.99%). Furniture sold more than Office Supplies (4.11 million against 3.79 million) but made a lot less profit (286,782 against 518,474),

because its margin was only 6.98%. So sales on their own don't tell you how a category is really doing.

6.2 Subcategories

Subcategory	Sales	Profit	Profit margin
Copiers	1,509,439	258,567.55	17.13%
Phones	1,706,874	216,717.01	12.70%
Paper	244,307	59,207.68	24.23%
Tables	757,034	−64,083.39	−8.47%

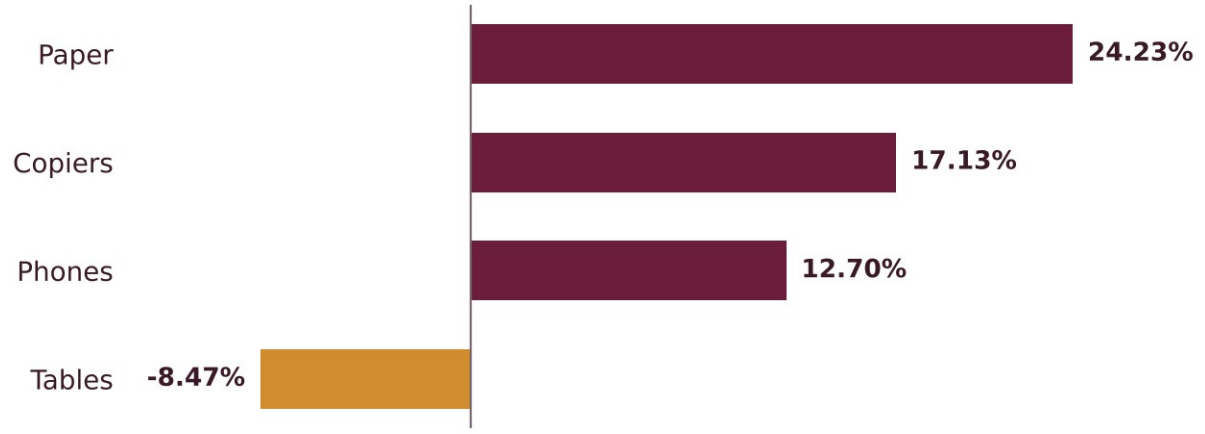


Figure 2. Profit margin for the subcategories reviewed

Copiers made the most profit of any subcategory. Paper had the highest margin, at 24.23%. Tables was the only subcategory with a loss, about 64,083.39 on sales of about 757,034.

A problem like Tables can get lost in the category totals, so it was worth looking one level down. These numbers show where to look, but not why Tables lost money.

6.3 Furniture and discounts

Furniture discount	Profit	Profit margin
0%	465,612.96	24.19%
20%	26,574.16	4.59%
30%	−20,011.89	−10.81%
40%	−61,464.32	−29.72%
50%	−47,347.33	−54.26%
70%	−67,210.92	−146.08%

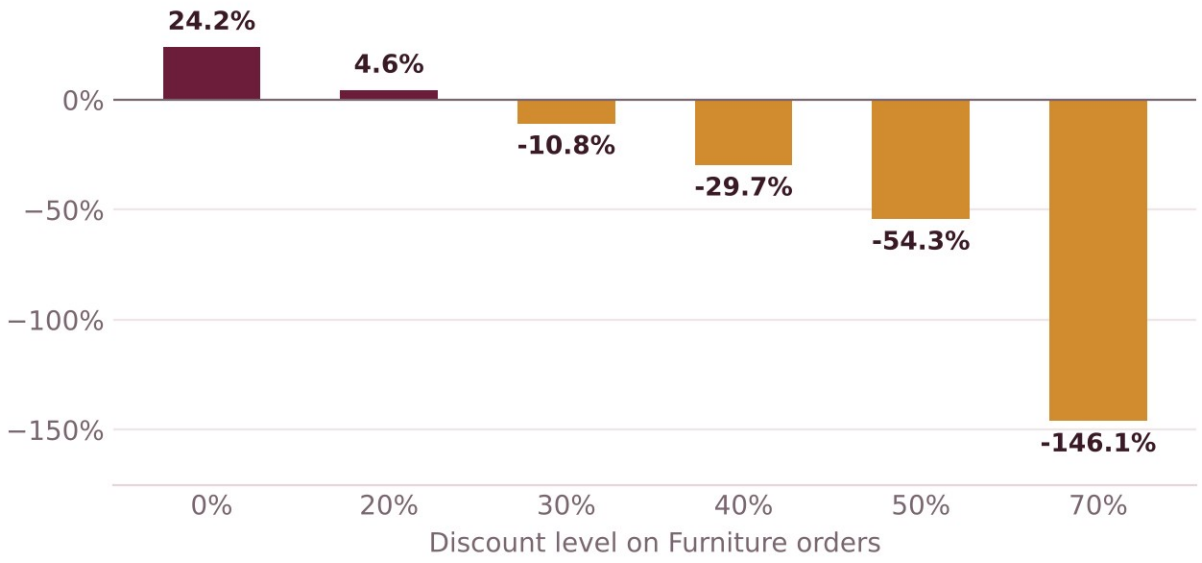


Figure 3. Furniture profit margin by discount level

With no discount, Furniture had a margin of 24.19%. At a 30% discount it was −10.81%, and at 40% it was −29.72%. Several discount groups from 30% upward lost money overall.

So in this dataset, higher discounts go together with lower profit. It doesn't show that the discounts caused it. Which products were discounted, what they cost and how big the orders were could explain some of the difference.

7. Recommendations

- Look at how Furniture discounts get approved, especially at the levels where margins went negative, and weigh the margin against the extra sales before approving big discounts.
- Look into Tables in more detail, by product, discount, region, customer segment and time period, to see where the losses are. Check costs, product mix and shipping too.

- Keep watching margin as well as sales. High sales don't always mean good profit, as Furniture and Tables show.
- Before changing any pricing or discount policy, compare similar products and orders, check how many orders sit behind each discount level, and look at costs.

8. Limitations

- This is an exploratory analysis. The link between discounts and profit is an association, and it doesn't show that discounts caused the losses.
- I only used the fields in the dataset, and the data covers 2011 to 2014, so it may not reflect how things look now or apply to other businesses.
- Some numeric fields were imported as text because of commas. My SQL removed the commas and converted the values before calculating.
- The discount results are totals for each group. Product mix, costs, order size, shipping and region may differ between groups.
- My Excel checks found no blanks in the fields I reviewed and no exact duplicate rows, but that doesn't guarantee every value is correct.
- The dashboard only covers category-level sales, profit and margin. Trends over time, regional differences and the causes of the Tables loss would need more analysis.

9. Files and dashboard

- Excel workbook with the initial data checks and date conversion
- SQL analysis in PostgreSQL / pgAdmin (KPIs, category, subcategory and discount comparisons)
- Tableau Public dashboard, "Sales & Profitability Analysis", with three category charts and a key finding

Dashboard:
https://public.tableau.com/views/SalesandProfitabilityAnalysis_17928000788560/Dashboard1

Appendix: Definitions

Measure	Definition
Total sales	Sum of sales across all records, after converting the text values to numbers.
Total profit	Sum of profit across all records, after converting the text values to numbers.
Profit margin	Total profit ÷ total sales × 100.
Category / subcategory margin	Profit margin worked out from the combined profit and sales of each category or subcategory.